PULSE Instrument Cluster

Requirements Traceability Matrix

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Contents

Document control

Field	Value
Document ID	PULSE-RTM-001
Title	Requirements Traceability Matrix
Generated	2026-09-03 by `scripts/gen_rtm.py`
Status	Derived; regenerate, never hand-edit
Sources	PULSE-StRS-001, PULSE-SyRS-001, PULSE-SAD-001, PULSE-SAF-001, PULSE-SEC-001

This matrix is produced from the ID tables in the source documents. A requirement's status, verification method and evidence are copied from its owning document. A story's trace list is the only link authored by hand. Dangling references fail the generator, so a matrix that exists is a matrix that is consistent.

1. Status summary

Group	Implemented	Partial	Planned S2	Planned S3	Planned S4	Total
Functional (FR)	16	4	0	1	1	22

Group	Implemented	Partial	Planned S2	Planned S3	Planned S4	Total
Interface (IR)	4	1	0	0	0	5
Non-functional (NFR)	4	2	0	2	0	8
Functional safety (SR)	4	4	1	1	1	11
Cybersecurity (CS)	3	4	1	2	1	11
All	**31**	**15**	**2**	**6**	**3**	**57**

Coverage: 36 of 57 requirements are reached by at least one user story; 21 are traced only through their owning document (listed in section 5).

2. Stakeholder needs to stories to requirements

Need	Statement	Stories	Requirements reached
SN-1	Decide the cluster toolkit on measured evidence, with application cost separated from platform cost	US-5, US-8, US-10	FR-7, FR-12, FR-13, FR-15, FR-16, FR-17, NFR-3, NFR-7
SN-2	Build in-house competence in the hybrid cockpit pattern: one broker, a Linux cluster, an Android surface	US-2, US-8	FR-2, FR-3, FR-12, FR-13
SN-3	An HMI decoupled from the vehicle bus so a platform change does not restart HMI work	US-1, US-3	FR-1, FR-5, FR-8, FR-11
SN-4	A demonstrator presentable to a non-technical audience in under five minutes	US-4, US-6, US-7, US-9	FR-6, FR-9, FR-10, FR-14, IR-3, IR-4, NFR-2
SN-5	Competence that outlives the project: a new engineer brings the stack up from documentation alone	US-11	NFR-5
SN-6	Compliance readiness: the safety and security artefacts a production programme would ask for, produced while building rather than retrofitt…	US-14, US-15, US-16, US-17	CS-1, CS-2, CS-3, CS-6, SR-1, SR-2, SR-3, SR-9, SR-11
SN-7	A closed loop, not a dashboard light: perception feeds a vehicle reaction through the same signal path the cluster renders	US-12, US-13	FR-18, FR-19, FR-20, FR-21, SR-7

3. Requirement to story, verification, status and evidence

3.1 Functional (FR), owner PULSE-SyRS-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
FR-1	The system shall represent all vehicle data using an open, standardised signal catalogue (COVESA VSS), not proprietary or bus-specific iden…	US-1	I	Implemented	`vss/vss_agl.json` compiled from VSS 6.0; 1,302 leaf signals

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
FR-2	The system shall extend the standard catalogue for domain-specific signals using the standard overlay mechanism only, with no forks of the…	US-2	I	Implemented	`vss/agl_vss_overlay.vspec`, `vss/pulse_vss_overlay.vspec`, `vss/dms_vss_overlay.vspec`; upstream spec consumed unmodif…
FR-3	The compiled signal tree shall be reproducible from versioned source files by a documented command.	US-2	T	Implemented	README "Regenerate the VSS JSON"; byte-identical regeneration checked by the regression suite (`vss-regeneration`), rep…
FR-4	A central broker shall serve the signal tree and allow multiple independent clients to subscribe concurrently.	none	D	Implemented	KUKSA databroker 0.7.0; Flutter cluster, Compose cluster and engine-audio service subscribe concurrently; suite check ...
FR-5	Signal producers and consumers shall be mutually decoupled: neither needs knowledge of the other's implementation.	US-3	I	Implemented	Providers and HMIs share no code, only VSS paths (PULSE-SAD-001 section 4)
FR-6	The system shall provide a telemetry source that generates physically plausible, correlated vehicle behaviour (speed, engine speed, gear, t…	US-4	D	Implemented	`emulator/telemetry_sim.py`: lap and cruise cycles, gear-dependent acceleration, RPM from speed and gear, fuel and odom…
FR-7	The telemetry source shall run a repeatable cycle so that UI behaviour can be compared across runs and across implementations.	US-5	T	Partial	Cycle is a fixed phase table with no randomness, but sampling is wall-clock driven so two runs differ in sample timing.…
FR-8	The telemetry source shall be replaceable by a real vehicle-bus provider without modification to any HMI code.	US-3	T	Partial	Met by design (HMIs subscribe to VSS paths only); verified when the virtual CAN feeder replaces the simulator in S3
FR-9	The HMI shall display, updating live: road speed, engine speed with warning and redline indication, selected gear, fuel level and range, pe…	US-6	D	Implemented	`pulse-cluster/lib/screen/pulse_screen.dart`; subscribed paths listed in PULSE-SAD-001 appendix A
FR-10	The HMI shall display motorsport telemetry: current lap number, current and best lap time, aerodynamic (DRS) state, and energy-recovery (ER…	US-7	D	Implemented	`Vehicle.Motorsport.*` overlay signals rendered by both HMIs

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
FR-11	The HMI shall subscribe to abstract signals only. It shall not contain CAN frame parsing, DBC knowledge, or hardware-specific decoding.	US-1	T	Implemented	Regression suite check `hmi-purity` scans both HMI sources for CAN, DBC and SocketCAN tokens; passing, report in `docs/...`
FR-12	The HMI shall be implemented twice, once in Flutter for Linux and once in Compose for Android Automotive, from a single shared visual desig…	US-8	I	Implemented	`pulse-cluster/` (Flutter) and `pulse-cluster-android/` (Compose); design source in `design/`
FR-13	Both implementations shall consume the identical signal source, so any observed difference is attributable to the platform, not the data.	US-8	D	Implemented	Both subscribe to the same broker on port 55555 via kuksa.val.v1
FR-14	The HMI shall render full-screen on the target strip display, and shall degrade gracefully to a normal window when that display is absent.	US-9	D	Implemented	`run.sh` locates a 2560 x 720 output at runtime and passes `CLUSTER_GEOM`; unset means a normal window
FR-15	The system shall support measurement of both implementations under identical telemetry: CPU utilisation, memory footprint, and shipped arte…	US-10	T	Partial	Measured once by hand (BRIEF.md key result: 123 MB vs 46 MB app memory, 12.2 % vs 11.5 % CPU, 50 MB vs 23 MB artefact).…
FR-16	Measurements shall distinguish application-level cost from platform-level cost, since these drive different procurement decisions.	US-10	A	Partial	Platform memory reported separately (AAOS 3.2 to 4.0 GB from emulator); target-hardware figures planned S3 to S4
FR-17	Results shall be documented such that a reviewer can reproduce them from the repository.	US-10	T	Planned S4	Benchmark script and results file to be committed under `docs/`
FR-18	The system shall derive driver fatigue and distraction levels from a driver-facing camera using inference executed entirely on the local de…	US-12	D	Implemented	`emulator/dms.py`: MediaPipe FaceLandmarker on CPU; publishes `Vehicle.Driver.FatigueLevel`, `DistractionLevel`, `Atten...`
FR-19	The system shall maintain a driver alert state (NONE, DROWSY, DISTRACTED, NO_FACE) with a configurable hold time before raising and clear t…	US-12	T	Implemented	`DMS_ALERT_HOLD_S` 3.0 s, `DMS_ALERT_CLEAR_S` 2.0 s; state published as `Vehicle.Driver.Monitoring.AlertState`
FR-20	When in cruise mode and the alert state is DROWSY, the vehicle model shall execute a minimum-risk manoeuvre: a countdown during which drive…	US-13	T	Implemented	`telemetry_sim.py` cruise mode: 5 s countdown, 22 km/h/s deceleration, resume after 5 s attentive; states published as…

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
FR-21	The HMI shall render the driver attention level, the active alert as a full-width banner distinguishable by colour per alert type, and the…	US-12	D	Implemented	Attention panel and banner in `pulse_screen.dart`; attention meter in `classic_screen.dart`
FR-22	The system shall accept an autonomy-stack provider that publishes planned trajectory, detected objects and drive mode into the broker as VS…	none	D	Planned S3	Autoware planning-simulation bridge, work package 04; VSS overlay for autonomy signals to be added under `vss/`

3.2 Interface (IR), owner PULSE-SyRS-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
IR-1	Client-to-broker communication shall use a language-neutral RPC interface supporting streaming subscriptions.	none	I	Implemented	gRPC over HTTP/2, `kuksa.val.v1` Subscribe stream; Dart and Kotlin stubs generated from the same `.proto`
IR-2	Signal semantics (units, ranges, allowed enumeration values, and encodings) shall be defined in the signal catalogue, not in client code.	none	I	Implemented	Units, `allowed` lists and `max` in the `.vspec` overlays; broker rejects out-of-catalogue values on set
IR-3	The target display interface shall be a 2560 x 720 automotive strip panel.	US-9	D	Implemented	Corsair Xeneon Edge on the desk rig; layout designed at 2560 x 720
IR-4	Display placement shall be configurable at runtime, not fixed at compilation.	US-9	D	Implemented	`CLUSTER_GEOM` environment variable read by the Linux runner; `run.sh` derives it from `xrandr`
IR-5	The system shall run without transport security in the local development configuration, and the security posture shall be an explicit, docu…	none	I	Partial	`--insecure` set in `scripts/start-databroker.sh`; documented as a deviation in PULSE-SEC-001 section 5. Start-up warni…

3.3 Non-functional (NFR), owner PULSE-SyRS-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
NFR-1	Responsiveness: displayed values shall track the telemetry source with no perceptible lag at a glance.	none	D	Implemented	Fast signals at 10 Hz, one broker tick from publish to render; observed at the review demo

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
NFR-2	Legibility: primary values (speed, gear, engine speed) shall be readable in a sub-second glance, consistent with driver-distraction practic…	US-6	D	Implemented	Design reviewed on the strip; numerals sized for the 720 px height
NFR-3	Footprint: the Linux implementation shall be deployable on constrained embedded hardware; total platform footprint is a first-class evaluat…	US-10	T	Planned S3	Raspberry Pi 5 baseline, work package 03, week 9
NFR-4	Portability: migration to a different SoC or to a Yocto/AGL image shall not require HMI rework.	none	A	Planned S3	Argument to be recorded with the Pi 5 baseline; Flutter embedder is the only platform-specific layer
NFR-5	Reproducibility: a new engineer shall be able to bring up the full stack on a clean machine from documentation alone.	US-11	T	Implemented	Clean-machine bring-up performed 2026-08-18 and recorded in ONBOARDING.md, six blockers closed
NFR-6	Openness: the stack shall be built from open-source components, with no proprietary runtime licence required for evaluation.	none	I	Implemented	KUKSA (Apache-2.0), VSS (MPL-2.0), Flutter (BSD), MediaPipe (Apache-2.0), AOSP; NOTICE file for the face model
NFR-7	Determinism: repeated runs of the same cycle shall produce the same signal sequence, so comparisons are valid.	US-5	T	Partial	Same as FR-7: value sequence is deterministic in shape, sample timing is not. Fixed-step replay planned S4
NFR-8	Maintainability: upstream components shall be consumed at pinned versions, not floating heads.	none	I	Partial	`kuksa-client==0.5.2` and Python deps pinned in `requirements.txt`; VSS pinned to v6.0; Dart deps locked in `pubspec.lo...`

3.4 Functional safety (SR), owner PULSE-SAF-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
SR-1	An item definition shall be written for the cluster and driver-monitoring function, naming the system boundary, the assumed vehicle context…	US-15	I	Partial	Section 2 of this document; not yet safety-reviewed

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
SR-2	A preliminary hazard analysis and risk assessment (HARA) shall be recorded for the displayed functions, with severity, exposure and control…	US-15	I	Partial	Section 4; eight hazards rated; review in S2
SR-3	Signals whose corruption or loss could mislead the driver (road speed, gear, warning indicators, drowsiness alert) shall be identified as s…	US-15	I	Implemented	Section 3
SR-4	The system shall detect loss of a safety-relevant signal, including broker disconnection and stale values, and shall indicate the degraded…	none	T	Planned S2	SM-1; Flutter service reconnects but shows last value today
SR-5	Every safety-relevant signal shall carry a freshness criterion; a value older than its criterion shall be treated as invalid.	none	T	Partial	Criteria proposed in section 3; enforcement with SM-1 in S2
SR-6	Safety-relevant display elements shall fail visibly. A rendering fault shall not present a plausible but wrong value.	none	T	Planned S3	SM-2
SR-7	The minimum-risk manoeuvre triggered by sustained driver drowsiness shall be specified as a safety mechanism: trigger condition, driver-rec…	US-13	D	Implemented	SM-3; implemented in `emulator/telemetry_sim.py` cruise mode; demonstrated live
SR-8	Freedom from interference shall be argued for the mixed-criticality cockpit: infotainment content shall not be able to degrade or obscure t…	none	A	Planned S4	Section 6 outline
SR-9	A safety-relevant requirement shall be traceable to the verification that demonstrates it, and each verification result shall be reproducib…	US-17	T	Partial	PULSE-RTM-001 generated from this table; regression suite runs before each demo and commits its report to `docs/evidenc...`
SR-10	Any decision that would block a later ASIL argument, such as the choice of graphics stack, operating system or toolkit, shall be recorded a…	none	I	Implemented	PULSE-SAD-001 section 8, DD-1 to DD-11
SR-11	A qualification gap list shall be maintained: for each component, what a production programme would still need (qualified toolchain, certif…	US-15	I	Implemented	Section 8

3.5 Cybersecurity (CS), owner PULSE-SEC-001

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
CS-1	A threat analysis and risk assessment (TARA) shall be recorded for the signal path, covering the broker, both HMI clients, the telemetry pr…	US-16	I	Partial	Section 4, nine scenarios; not yet security-reviewed
CS-2	The item boundary and its trust boundaries shall be documented, distinguishing on-device interfaces from anything reachable off the device.	US-16	I	Implemented	Section 2
CS-3	Broker communication shall support authenticated, encrypted transport (mutual TLS). Running without it shall be a configuration reserved fo…	US-16	T	Planned S2	Section 5; `--insecure` in `scripts/start-databroker.sh` today
CS-4	Write access to the signal tree shall be authorised per client. A consumer shall not be able to publish, and no client shall write outside…	none	T	Planned S3	Section 5
CS-5	Signal values received from any provider shall be range- and type-validated against the catalogue before display, so a compromised provider…	none	T	Partial	Broker-side validation implemented by KUKSA; HMI-side check and test in S2
CS-6	Camera frames shall not leave the device and shall not be persisted. Only derived state, such as the drowsiness level, shall be published.	US-14	I	Implemented	`emulator/dms.py` publishes six derived signals and writes no files; test in S2
CS-7	Security-relevant events, including authentication failure, unauthorised write attempt and client disconnection, shall be logged with a tim…	none	T	Planned S3	Section 5
CS-8	All third-party dependencies shall be pinned and inventoried as a software bill of materials, and the inventory shall be checkable against…	none	T	Partial	Python and Dart pinned; container image at `latest`; no SBOM. S2
CS-9	No credential, key or certificate shall be committed to the repository; the development configuration shall use clearly marked non-producti…	none	T	Implemented	Suite check `secrets` scans every tracked file for key and credential patterns; passing

ID	Requirement	Stories	Verif.	Status (W3)	Evidence
CS-10	An update concept shall be described for cluster and monitor software, covering authenticity of the package and behaviour on a failed updat…	none	I	Planned S4	Section 6 outline
CS-11	Cybersecurity requirements shall be verified as part of the sprint regression suite, not as a separate end-of-project activity.	none	T	Partial	Suite exists (`scripts/regression.sh`) with range and secret checks for CS-5 and CS-9; CS-6 and CS-8 checks join in S2

4. Story to requirements

Story	Statement	Needs	Requirements	Verification of those requirements
US-1	As an HMI developer, I want every vehicle value available as a named VSS signal, so that I never touch bus formats or h…	SN-3	FR-1, FR-11	I, T
US-2	As a platform engineer, I want domain signals added as VSS overlays, so that we stay compatible with the upstream catal…	SN-2	FR-2, FR-3	I, T
US-3	As an integrator, I want producers and consumers to meet only at the broker, so that either side can be swapped without…	SN-3	FR-5, FR-8	I, T
US-4	As a demonstrator, I want a realistic drive cycle without a vehicle, so that the cluster behaves plausibly on a desk.	SN-4	FR-6	D
US-5	As an evaluator, I want the same cycle to replay identically, so that measurements of the two HMIs are comparable.	SN-1	FR-7, NFR-7	T
US-6	As a driver, I want speed, RPM, gear, fuel and tyre pressures at a glance, so that I can read the vehicle state in unde…	SN-4	FR-9, NFR-2	D
US-7	As a race engineer, I want lap, DRS and ERS state on the cluster, so that motorsport telemetry is visible live.	SN-4	FR-10	D
US-8	As a stakeholder, I want the same design built in Flutter and Compose from one signal source, so that the platform comp…	SN-1, SN-2	FR-12, FR-13	D, I
US-9	As a presenter, I want the cluster full-screen on the strip display but windowed elsewhere, so that the demo works on a…	SN-4	FR-14, IR-3, IR-4	D

Story	Statement	Needs	Requirements	Verification of those requirements
US-10	As the sponsor, I want application cost separated from platform cost, so that the toolkit decision and the hardware bil…	SN-1	FR-15, FR-16, FR-17, NFR-3	A, T
US-11	As a new engineer, I want to bring the stack up from documentation alone, so that the competence outlives the project.	SN-5	NFR-5	T
US-12	As a driver, I want the cluster to warn me when I am drowsy or distracted, so that I regain attention before anything e…	SN-7	FR-18, FR-19, FR-21	D, T
US-13	As a safety engineer, I want sustained drowsiness to trigger a defined minimum-risk manoeuvre with a recovery window, s…	SN-7	FR-20, SR-7	D, T
US-14	As a privacy officer, I want camera frames to stay on the device, so that driver monitoring never becomes surveillance.	SN-6	CS-6	I
US-15	As a safety manager, I want an item definition, hazard analysis and safety-relevant signal list for the cluster, so tha…	SN-6	SR-1, SR-2, SR-3, SR-11	I
US-16	As a security manager, I want a threat analysis and trust-boundary description for the signal path, so that the desk-on…	SN-6	CS-1, CS-2, CS-3	I, T
US-17	As a reviewer, I want every requirement traced to a story and a verification, so that I can see what the prototype cove…	SN-6	SR-9	T

5. Requirements not reached by a user story

These are traced only through their owning document and its verification column. Add a story or accept the gap at review.

ID	Requirement	Owner	Status (W3)
CS-4	Write access to the signal tree shall be authorised per client. A consumer shall not be able to publish, and no client shall write outside…	PULSE-SEC-001	Planned S3
CS-5	Signal values received from any provider shall be range- and type-validated against the catalogue before display, so a compromised provider…	PULSE-SEC-001	Partial
CS-7	Security-relevant events, including authentication failure, unauthorised write attempt and client disconnection, shall be logged with a tim…	PULSE-SEC-001	Planned S3
CS-8	All third-party dependencies shall be pinned and inventoried as a software bill of materials, and the inventory shall be checkable against…	PULSE-SEC-001	Partial
CS-9	No credential, key or certificate shall be committed to the repository; the development configuration shall use clearly marked non-producti…	PULSE-SEC-001	Implemented

ID	Requirement	Owner	Status (W3)
CS-10	An update concept shall be described for cluster and monitor software, covering authenticity of the package and behaviour on a failed updat…	PULSE-SEC-001	Planned S4
CS-11	Cybersecurity requirements shall be verified as part of the sprint regression suite, not as a separate end-of-project activity.	PULSE-SEC-001	Partial
FR-4	A central broker shall serve the signal tree and allow multiple independent clients to subscribe concurrently.	PULSE-SyRS-001	Implemented
FR-22	The system shall accept an autonomy-stack provider that publishes planned trajectory, detected objects and drive mode into the broker as VS…	PULSE-SyRS-001	Planned S3
IR-1	Client-to-broker communication shall use a language-neutral RPC interface supporting streaming subscriptions.	PULSE-SyRS-001	Implemented
IR-2	Signal semantics (units, ranges, allowed enumeration values, and encodings) shall be defined in the signal catalogue, not in client code.	PULSE-SyRS-001	Implemented
IR-5	The system shall run without transport security in the local development configuration, and the security posture shall be an explicit, docu…	PULSE-SyRS-001	Partial
NFR-1	Responsiveness: displayed values shall track the telemetry source with no perceptible lag at a glance.	PULSE-SyRS-001	Implemented
NFR-4	Portability: migration to a different SoC or to a Yocto/AGL image shall not require HMI rework.	PULSE-SyRS-001	Planned S3
NFR-6	Openness: the stack shall be built from open-source components, with no proprietary runtime licence required for evaluation.	PULSE-SyRS-001	Implemented
NFR-8	Maintainability: upstream components shall be consumed at pinned versions, not floating heads.	PULSE-SyRS-001	Partial
SR-4	The system shall detect loss of a safety-relevant signal, including broker disconnection and stale values, and shall indicate the degraded…	PULSE-SAF-001	Planned S2
SR-5	Every safety-relevant signal shall carry a freshness criterion; a value older than its criterion shall be treated as invalid.	PULSE-SAF-001	Partial
SR-6	Safety-relevant display elements shall fail visibly. A rendering fault shall not present a plausible but wrong value.	PULSE-SAF-001	Planned S3
SR-8	Freedom from interference shall be argued for the mixed-criticality cockpit: infotainment content shall not be able to degrade or obscure t…	PULSE-SAF-001	Planned S4
SR-10	Any decision that would block a later ASIL argument, such as the choice of graphics stack, operating system or toolkit, shall be recorded a…	PULSE-SAF-001	Implemented

6. Consistency

Needs: 7. Stories: 17. Requirements: 57. Dangling references: 0.